INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Business Continuity and Disaster Recovery Assignment

GRC205 Incident Management and Business Continuity – Week 21 | Northbridge Skills Institute (fictional)

Item	Detail
Assessment	GRC205 Business Continuity and Disaster Recovery Assignment (Week 21) – 100 marks, 30% of course grade
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Reading guide

The six tasks follow in order, with calculations and matrices in tables and the reasoning in the text. The Task 5 schedule gives **modelled baseline results**, not a live test, and the tabletop is a **simulation**. All data are fictional. Week 21 slide citations refer to the instructor's teaching deck (ICDFA, n.d.). Table 1 lists my assumptions (A1–A12).

Table 1. Assumptions (not stated in the brief).

ID	Assumption	Where used
A1	The examination lasts 60 minutes, with 30 minutes between sittings for changeover and identity checks. The brief gives no duration.	Tasks 4, 5
A2	Microsoft 365 (already available through mobile internet) includes a forms or quiz tool that the Head of Examinations can approve as a temporary assessment service, and the approved question set is held in M365 or in an offline copy outside the flooded site. Unverified: if false, the NGN 120,000 rescheduling cost applies.	Tasks 1, 4
A3	DB01 incremental backups run hourly on the hour. The brief states only that the chain runs to 07:00.	Tasks 2, 5
A4	The 40 learners needing a centre workstation and the 80 using their own devices are the same 120 learners scheduled for the 11:00 examination.	Tasks 4, 5
A5	The offsite provider copies can be retrieved and decrypted, and the encryption keys are not held only at the flooded site. Unverified.	Tasks 2, 4
A6	Hourly disruption costs stop for a function when its application or file service has been restored, not when the business accepts it. This is a modelling simplification.	Tasks 3, 5
A7	The Hot package restores all five functions in about one hour and the Warm package follows the current drill timings. Package claims are planning estimates.	Task 3
A8	The primary site is flood-prone and the alternate facility is outside the same flood zone. Unverified.	Tasks 3, 6
A9	The exercise is run on 23 October 2026 for dating corrective actions.	Task 5
A10	The Director is the authorised decision maker, and the Registrar is the deputy.	Task 4
A11	The incident response (IR) plan prepared for the Week 20 assignment is the organisation's IR plan, so the Incident Response Lead (IRL) role is already defined.	Task 4
A12	Payment-provider records are the authoritative record of payments received while FIN01 is down.	Tasks 4, 5

Task 1 – Business Impact Analysis (20 marks)

1.1 Five-function BIA

The BIA covers the five functions that the main-site outage stops. For each, Table 2 records the owner, users, resources, dependencies and minimum acceptable service, with the people, premises, supplier and information needs. Minimum

service is the least that the business can accept without unmanaged harm; where it relies on a manual process, I name the approval and record controls (NIST, 2010, BIA step; ICDFA, n.d., slide 5).

Table 2. Business Impact Analysis, F1–F5 (impact over time uses the supplied NGN-per-hour rates).

	F1 Teaching and assessment	F2 Fees and finance	F3 Admissions and registration	F4 Staff payroll	F5 Records and certificates
Owner role	Head of Examinations	Finance Manager	Admissions Manager	HR and Payroll Manager	Registrar
Users affected	1,800 learners (120 in 11:00 exam), teaching staff	Learners paying fees, finance staff, 90 staff for receipts	Applicants, admissions staff, learners changing registration	90 staff	Learners and alumni requesting records, records staff
Critical resources (systems)	LMS01, DB01	FIN01, DB01	REG01, DB01	HR01, DB01	FS01
Shared dependencies	Recovery network, AUTH01, secure access, communication, staff	Same, plus payment provider portal	Same	Same, plus bank submission channel	Network, AUTH01, FS01, staff
People	Invigilators, 2 technicians, business coordinators	Finance officers (dual control)	Admissions staff	HR and finance approvers	Records officers
Premises	Alternate facility (25 workstations from 10:00), learners' own devices	Remote work or alternate facility	Remote work	Remote work	Remote work
Suppliers	Alternate facility, ISP and mobile data, Microsoft	External payment provider	Microsoft (online forms)	Bank	Offsite backup provider
Information needs	Approved question set, learner list, accessibility needs	Provider transaction reports, fee ledger	Applications received, closing list	Last approved payroll, approved changes	Request log, access list
Impact over time	Exam missed at 11:00 (3 h): NGN 120,000 one-off. NGN 60,000/h; learner access and assessment integrity at risk	Fee confirmation due 16:00 (8 h). NGN 40,000/h; unreconciled receipts	Admissions close 18:00 (10 h). NGN 20,000/h; lost applicants	Bank file due 12:00 Day 1 (28 h). NGN 10,000/h; wrong or duplicate pay	48 h service commitment. NGN 5,000/h; integrity and access control risk
Minimum acceptable service	Authenticated exam delivery via temporary tool with invigilation; read access to course materials	Manual receipt register using provider reports, dual control; no wallet edits	Online application form with restricted list; closing time honoured or formally extended	Last approved payroll plus dual-approved changes only; single bank file	Request queue logged; urgent issues signed by Registrar after verification
Priority	1	2	3	4	5

1.2 Worked impact calculations

The hourly figures are separate, non-overlapping incremental costs, so they add. The combined rate is 60,000 + 40,000 + 20,000 + 10,000 + 5,000 = NGN 135,000 per hour. With no recovery or continuity measures the examination at 11:00 is missed in both periods, so the one-off NGN 120,000 is added once.

Table 3. Disruption cost with no measures (NGN). Cost = rate × hours; the exam trigger is added once.

Function	Rate/h	8 h calculation	8 h cost	24 h calculation	24 h cost
F1 Teaching and assessment	60,000	60,000 × 8	480,000	60,000 × 24	1,440,000
F2 Fees and finance	40,000	40,000 × 8	320,000	40,000 × 24	960,000
F3 Admissions and registration	20,000	20,000 × 8	160,000	20,000 × 24	480,000
F4 Staff payroll	10,000	10,000 × 8	80,000	10,000 × 24	240,000
F5 Records and certificates	5,000	5,000 × 8	40,000	5,000 × 24	120,000
Subtotal (hourly)	135,000	135,000 × 8	1,080,000	135,000 × 24	3,240,000
Examination rescheduling (one-off)	–	Trigger met (no service by 11:00)	120,000	Trigger met	120,000
Total estimated disruption cost		1,080,000 + 120,000	**1,200,000**	3,240,000 + 120,000	**3,360,000**

Limitations. First, the rates are exercise estimates and are linear, whereas real harm is uneven: a missed payroll or fee deadline costs far more than an average hour, and the first hours of a quiet night cost less. Second, the figures include only the supplied incremental costs and omit recovery spending, overtime, lost future enrolment and contractual penalties, so they neither measure nor cap the true loss. Financial impact must also be kept apart from learner welfare (120 learners' examination and assessment fairness), reputation, and **data integrity** (duplicate payroll, lost receipts, altered records), none of which the hourly rates price. I therefore treat integrity and welfare as constraints on how fast I recover, not as items to trade for money.

1.3 Priorities and dependency risks

Business priority follows deadline and cost: F1 first because the 11:00 examination is 3 hours away and the rate is highest, then F2 (16:00, 40,000/h), F3 (18:00), F4 (12:00 Day 1) and F5 (48 hours). **Technical restoration order** is different because the systems share infrastructure (ICDFA, n.d., slide 5): the recovery network, AUTH01, and the security and backup checks must precede DB01, and every application needs DB01 (F5 needs only AUTH01 and FS01). The order is therefore: network, AUTH01, security and backup checks, DB01, then LMS01, FIN01, REG01, HR01 in business order, with FS01 restored on the second technician during the DB01 restore because it does not delay the critical path.

Restoration alone cannot protect the 11:00 examination: Task 5 shows the fastest assessment service is 11:45, so the examination gets a continuity workaround (Task 4) and DB01 restoration serves the later deadlines.

Dependency risks:

- **DB01 is a single point of failure.** F1–F4 all wait for its 90-minute restore, and one corrupt incremental stops them all.
- **Identity split.** Microsoft 365 is available but local applications use AUTH01, whose snapshot is 8 hours old, so accounts or passwords changed since 00:00 may not match.
- **Shared administrator identity.** Production and backup administration share one identity, so one compromise could damage both, and the offsite provider's access independence is undocumented.
- **Two technicians.** Only two people can run actions, fatigue grows over 72 hours (or longer), and an absence halves capacity.
- **Alternate facility capacity and mobile internet.** 25 workstations serve 40 learners who need them, and M365 reaches everyone only through mobile data.

Task 2 – Recovery objectives and backup design (15 marks)

2.1 Objectives and recovery declaration

Recovery Time Objective (RTO) is the target time from disruption (08:00) to restoring a function at its agreed minimum service. **Recovery Point Objective (RPO)** is the maximum age of recoverable data, the data loss I accept. **Maximum Tolerable Downtime (MTD)** is the point where disruption becomes unacceptable to the business, normally a deadline in the brief (NIST, 2010; ICDFA, n.d., slide 6). RTO must sit inside MTD with a buffer, because the first attempt can fail.

My rule is RTO ≤ 60% of MTD, leaving at least 40% of the tolerable time for failed steps, business acceptance and manual workarounds. MTDs come from deadlines: F2 16:00 (8 h), F3 18:00 (10 h), F4 12:00 Day 1 (28 h), F5 the 48-hour commitment. For F1 I use 8 hours (the end of the working day), because the single 11:00 examination (3 hours) cannot be guaranteed by restoration; it is covered by a workaround in Task 4. RTOs are checked against the current drill: LMS01 is available at 3 h 45 min, so a 4-hour target is achievable but has only a 15-minute margin, and I will not promise a shorter one until a test shows it.

Table 4. Objective matrix (all times from 08:00 Day 0).

Function	MTD	RTO	Buffer (MTD – RTO)	RTO ÷ MTD	RPO	Recovery declared when
F1 Teaching and assessment	8 h (16:00)	4 h (12:00)	4 h	50%	1 h	LMS01 login works for the exam cohort; a test submission is saved and read back; integrity check passed; Head of Examinations signs
F2 Fees and finance	8 h (16:00)	4 h 30 min (12:30)	3 h 30 min	56%	1 h	FIN01 posts a test receipt; ledger agrees with the provider report to 07:00; Finance Manager signs

Function	MTD	RTO	Buffer (MTD – RTO)	RTO ÷ MTD	RPO	Recovery declared when
F3 Admissions and registration	10 h (18:00)	6 h (14:00)	4 h	60%	1 h	REG01 accepts and retrieves a test application; Admissions Manager signs
F4 Staff payroll	28 h (12:00 Day 1)	12 h (20:00)	16 h	43%	4 h	HR01 payroll totals match the last approved run; dual approval of any changes
F5 Records and certificates	48 h (08:00 Day 2)	24 h (08:00 Day 1)	24 h	50%	4 h	FS01 files open, permissions match AUTH01 groups; Registrar signs
Shared: network / AUTH01 / DB01	–	30 min / 1 h / 3 h	–	–	– / 4 h / 1 h	Each passes its own validation before dependants start

Recovery is **declared** at the minimum-service level in the last column, not when the technology starts. Restoring LMS01 is technical recovery; recovery is declared only after business acceptance, because a restored system that holds stale or corrupt data does not resume the function.

2.2 Checkpoint ages and data-loss windows

Using 08:00 on Day 0 as the reference point, the age of each checkpoint is the time between the checkpoint and 08:00.

Table 5. Checkpoint calculations and gaps against proposed RPOs.

Item	Checkpoint	Calculation	Age at 08:00	Potential data-loss window	Functions affected	Proposed RPO	Gap
B02 DB01 (B01 + incrementals)	07:00 Day 0	08:00 − 07:00	1 h	07:00–08:00 Day 0	F1, F2, F3, F4	F1–F3: 1 h; F4: 4 h	None (F1–F3 at limit, no margin)
B01 DB01 full (fallback base)	23:00 Day −1	24:00 − 23:00 + 08:00 = 1 + 8	9 h	23:00–08:00	F1–F4 if chain breaks	1 h / 4 h	8 h over F1–F3; 5 h over F4
B03 AUTH01 snapshot	00:00 Day 0	08:00 − 00:00	8 h	00:00–08:00 (accounts, groups, settings)	F1–F5	AUTH01: 4 h	4 h over
B04 FS01 full	23:00 Day −1	1 + 8	9 h	23:00–08:00	F5	4 h	5 h over
Disconnected full	17:00 Day −3	3 × 24 − 9 = 72 − 9	63 h	17:00 Day −3 to 08:00	Last-resort fallback	–	Not usable for RPO

DB01 meets its RPOs only if every incremental from B01 to 07:00 passes its check, which is why Task 5 stresses that assumption. AUTH01 (4 h over) and FS01 (5 h over) fail the proposed RPOs, so the backup design below closes those gaps.

What can be reconciled, and what cannot be assumed. Payments from 07:00 can be rebuilt from the payment provider's transaction report (A12). Applications or requests received by email or online forms can be re-entered. M365 audit logs and helpdesk tickets can show account changes since 00:00. I cannot assume that anything held only on local disks (receipts or wallet entries from 07:00 to 08:00, new FS01 files since 23:00, AUTH01 changes) is recoverable, so staff must declare these and I treat them as lost until found.

2.3 Backup and restoration design

The design closes the RPO gaps, applies the 3-2-1 rule (three copies, two media types, one offsite) and adds an independent immutable or offline copy that nobody with production administrator access can alter (CISA et al., 2023). The 400,000 add-on in Task 3 pays for the independent copy.

Table 6. Proposed backup schedule.

Asset	Full	Differential	Incremental	Retention	Copies and placement
DB01 (F1–F4)	Daily 23:00	04:00 and 12:00	Hourly on the hour	14 daily, 8 weekly, 12 monthly (confirm legal retention)	Copy 1 production; Copy 2 local backup repository (separate disk); Copy 3 encrypted offsite within 15 minutes, in a region outside the flood zone; Copy 4 immutable (object-lock, 30 days) under a separate identity
AUTH01	Snapshot every 4 h and after any change	–	–	14 days	As above, plus an exported configuration file stored offline

Asset	Full	Differential	Incremental	Retention	Copies and placement
FS01 (F5)	Daily 23:00	–	Every 4 h	30 daily, 12 monthly	As above; file versioning on
LMS01, FIN01, REG01, HR01 (application images)	Weekly and after upgrade	–	–	8 weeks	Offsite copy plus offline copy

Exact DB01 restore chain. Today, to reach 07:00: B01 (23:00 full) plus every hourly incremental at 00:00, 01:00, 02:00, 03:00, 04:00, 05:00, 06:00 and 07:00 (A3), eight incrementals in sequence. One failed file stops the chain at the point before it. With the proposed schedule, the chain to 07:00 is B01, the 04:00 differential, then incrementals 05:00, 06:00 and 07:00, which is five pieces and a fallback to 04:00 or to B01 even if a later incremental fails.

Security controls. Production and backup administration will use **separate identities** with MFA, no standing privilege, and alerting on every backup-administrator logon. Backups are encrypted (AES-256 at rest, TLS in transit); two key custodians hold the keys in a vault that is not only at the flooded site (A5). Restoration is tested monthly (sample restore of DB01 to a test server), quarterly (full DB01 and application restore) and yearly (full failover).

Why offsite storage or a replica is not proof of recoverability. An offsite copy proves only that data was copied. It may be incomplete, corrupt, encrypted by attackers, or unreadable without keys, and a live replica copies deletions, corruption and malicious changes within seconds, so it can destroy both sides together. Recovery is proved only when a restore from that copy has produced usable data inside the target time.

Table 7. Verification of integrity, access independence and usability.

Check	Method	Frequency	Pass criterion	Owner
Backup integrity	Checksum on creation and at the offsite copy; list the chain and confirm no gap; catalogue comparison	Every backup; chain check nightly	100% of files verified; no gaps from base to latest point	Technical Recovery Lead
Access independence	Review the provider's identity and access settings; test that production administrator credentials cannot delete or change the offsite or immutable copy; remove the shared identity	Quarterly and after any staff change	Delete attempt fails and is logged; no shared identity remains	BC/DR Coordinator
Usability of restored data	Restore to an isolated network; database consistency check; row counts and control totals against the last production report; application smoke test; function owner signs	Monthly sample; quarterly full	Totals agree; test transaction succeeds; owner sign-off recorded	Function owners
Unverified assumptions	A3 (hourly cadence), A5 (provider retrieval and keys), provider independence (undocumented)	Before 30 days	Each confirmed in writing or replaced by a tested alternative	BC/DR Coordinator

Task 3 – Recovery strategy and budget (15 marks)

3.1 Option comparison

I compared the three packages with the Task 2 targets (F1 RTO 4 h, F2 4 h 30 min, F3 6 h, F4 12 h, F5 24 h; RPO 1 h for F1–F3). The package timings are planning estimates that I must validate, so I do not treat any of them as a guarantee (ICDFA, n.d., slides 7 and 15).

Table 8. Option matrix against objectives and risks.

Criterion	Cold	Warm	Hot
First critical service (estimate)	24–48 h	3–5 h (drill: 3 h 45 min)	About 1 h
Data-loss window (estimate)	Up to 24 h	1–6 h	Up to 15 min
Meets RTOs (F1–F5)	No: fails F1–F4; F5 only at the 24 h edge	Mostly: F1 (4 h) at 3–4 h but not at 5 h; F2–F5 yes	Yes, with large margin
Meets RPO 1 h (F1–F3)	No (24 h)	Only at the 1 h end; needs the Task 2 backup schedule to hold the claim	Yes (15 min)
Staffing (2 technicians)	Very high manual build; rebuild for 24–48 h under fatigue	Fits the drill: 2 technicians for about 4 h 15 min	Needs less manual work but continuous monitoring and failover skills
Geography	Depends on where hardware is delivered	Alternate site must be outside the flood zone (A8)	Second region; network path must be diverse
Supplier dependencies	Hardware delivery time	Alternate facility, provider restore support	Replication provider, link provider, licences

Criterion	Cold	Warm	Hot
Corruption or malicious-change risk	Low propagation, but old data	Moderate: restore point can be chosen	High: live replica copies bad changes within seconds; independent backups still needed

3.2 Itemised annual budget

Every package carries two mandatory additions, NGN 400,000 for independent protected backups and NGN 200,000 for training and tests, a combined NGN 600,000. The annual limit is NGN 3,000,000.

Table 9. Itemised annual cost (NGN) against the NGN 3,000,000 limit.

Package	Base	Independent protected backups	Training and tests	Total	Share of limit	Headroom	Fits limit?
Cold	600,000	400,000	200,000	**1,200,000**	40%	1,800,000	Yes
Warm	1,500,000	400,000	200,000	**2,100,000**	70%	900,000	Yes
Hot	2,400,000	400,000	200,000	**3,000,000**	100%	0	Yes, exactly at the limit; no contingency

Totals: Cold 600,000 + 400,000 + 200,000 = 1,200,000; Warm 1,500,000 + 600,000 = 2,100,000; Hot 2,400,000 + 600,000 = 3,000,000. All three fit, but Hot leaves nothing for price changes, extra testing or a fault.

3.3 Recommendation

I recommend the Warm package at NGN 2,100,000. It is the lowest option that can meet F2–F5 and the 1-hour DB01 data-loss target with evidence, it fits the drill and the two-technician team, and it leaves NGN 900,000 for validation and contingency. Cold cannot meet any short RTO or the 1-hour RPO. Hot meets everything on paper, but it uses the full limit, is exposed to propagated corruption unless the independent backups work, and its claim of an hour is as unproven as Warm's.

What Warm cannot meet. It cannot guarantee LMS01 by 11:00: the drill gives 11:45 and even the best estimate (3 hours) has no margin. The examination is therefore protected by the continuity workaround, not by recovery. F1's 4-hour RTO has a 15-minute margin, and the 1-hour data-loss window relies on the redesigned backups.

Table 10. Residual risks and who must accept them.

Residual risk	Treatment	Accepted by
Examination at 11:00 cannot be met by restoration; depends on the temporary tool (A2)	Workaround approved in advance; mock exam tested; NGN 120,000 if it fails	Director
F1 RTO margin is 15 minutes; one failed step breaches it	Pre-stage network and AUTH01 images; target 3 h (Task 5 actions)	Director and Registrar
Data loss up to 1 h (DB01), 4 h (AUTH01, FS01) and any local-only records	Reconciliation from provider reports and staff declarations	Function owners
Supplier recovery claims unproven; alternate facility capacity (25) unverified	Contract addendum and timed acceptance test before approval	Director
Shared administrator identity until separated; provider access independence undocumented	Separate identities within 14 days; documented provider review	Technical Recovery Lead (risk owner); Director (acceptance of delay)
Two-technician single point of failure	Cross-training and vendor support	Director

3.4 Comparison with disruption costs and investment reasoning

Warm costs NGN 2.1 million a year, which is 1.75 times the 8-hour no-measures estimate (NGN 1.2 million) and 62.5% of the 24-hour estimate (NGN 3.36 million). I modelled each package's cost for one outage by applying the hourly rates until each function is restored (A6): Warm 225,000 + 170,000 + 70,000 + 42,500 + 12,500 = NGN 520,000 from the drill timings in Task 5; Cold at its best case of 24 hours, 135,000 × 24 = NGN 3,240,000; Hot at about one hour, NGN 135,000. The examination trigger is left out of all three because the workaround applies whichever package is bought.

Table 11. One-outage comparison and break-even frequency (illustrative; costs in NGN).

Comparison	Extra annual cost	Cost avoided per outage	Calculation	Break-even outages per year
Warm instead of Cold	900,000	2,720,000	3,240,000 − 520,000	0.33 (about one in three years)
Hot instead of Warm	900,000	385,000	520,000 − 135,000	2.34 outages a year

These numbers show a sensitivity, not a return on investment. A single-outage estimate has no probability, so it cannot give the annual expected saving, which needs the frequency and duration of floods and power failures and the effectiveness of each option. It also leaves out harm that the rates do not price. The table does say what must be believed: Warm beats Cold if a comparable outage happens more often than about once in three years, whereas Hot beats Warm only if more than two happen every year, which I do not expect.

Evidence needed before approving the spend or accepting supplier claims: the site's flood and outage history; written service levels with penalties and references; a timed proof-of-concept restore of DB01 and LMS01; an on-site test of 25 working workstations; backup-provider access independence; written prices within the limit; and the exam board's acceptance of the temporary tool.

Task 4 – Business continuity and disaster recovery plan (20 marks)

4.1 Coordinated plan: scope, activation and teams

Scope. The plan covers the main site, its local systems (LMS01, FIN01, REG01, HR01, FS01, DB01, AUTH01), the five functions and the people and suppliers they need, for outages up to the five-day stress case in Task 5. It is activated when any trigger is met: the site or a critical system is expected to be unavailable for more than 4 hours; a function deadline within 8 hours is at risk; AUTH01 or DB01 is lost; or life-safety or facility closure occurs. The 08:00 outage meets all four.

Authority and escalation. The Director is the authorised decision maker and the Registrar the deputy if the Director cannot be reached within 15 minutes (A10). I, as BC/DR Coordinator, can activate the plan on the thresholds above and the Director confirms within 30 minutes. Escalation runs from function owner to me, then to the Director, with an immediate call to the Incident Response Lead (IRL) for any suspected compromise. Contact roles (Director, Registrar, coordinator, technical lead, Facilities, Communications, IRL, function owners, suppliers) are on a printed and an offline call tree.

Alternative communication and reporting. Voice calls and SMS are primary because M365 relies on mobile data; M365 Teams or email is secondary, with no sensitive details by email until IR clears it. Situation reports go to the Director every 2 hours for the first 12 hours (10:00, 12:00, 14:00 and so on), then every 4 hours: the status of each function against its RTO and MTD, issues, decisions needed and the next report time.

How the teams work together. The continuity team (business coordinators and function owners) runs manual procedures and learner support. The technical recovery team (two technicians) runs the runbook. Facilities manages the site, power, alternate facility access and safety. The IR team (the IRL) joins when compromise is possible. I coordinate the four, hold the single decision log, and decide when technology is restored versus when a service has resumed (NIST, 2010; ICDFA, n.d., slides 12 and 14).

4.2 RACI matrix

R is Responsible, A is Accountable, C is Consulted and I is Informed. Each row has exactly one A and at least one R.

Table 12. RACI (A/R = one role both accountable and responsible).

Activity	DIR	BCDRC	TRL	BC	FO	FAC	IRL	COM
Plan activation	A	R	C	I	I	C	I	I
Temporary assessment delivery	I	C	C	R	A	C	I	C
Backup selection and validation	I	C	A/R	–	I	–	C	–
System restoration (runbook)	I	C	A/R	–	I	C	C	–
Business acceptance and service resumption	I	R	C	I	A	–	I	I

Activity	DIR	BCDRC	TRL	BC	FO	FAC	IRL	COM
Stakeholder communication	C	C	I	R	C	I	I	A
Incident response escalation	I	C	R	–	I	–	A	I
Facilities clearance and site safety	I	C	I	–	I	A/R	–	I
Return to the primary site	A	R	R	C	C	R	C	I
Situation reporting	I	A/R	C	R	C	C	I	I

Key: DIR Director; BCDRC BC/DR Coordinator; TRL Technical Recovery Lead (senior technician); BC business coordinators (continuity team); FO function owner (Head of Examinations for F1); FAC Facilities Manager; IRL Incident Response Lead; COM Communications Lead.

4.3 Temporary procedures for F1–F5 (72-hour outage)

The temporary procedures keep each function at its minimum service until its system is accepted. The alternate facility offers 25 workstations from 10:00, but 40 of the 120 examination learners need one, so I run two sittings: sitting 1 at 11:00 for 25 centre learners and the 80 learners using their own devices, and sitting 2 at 12:30 for the remaining 15 (A1, A4). Learners with accessibility needs are placed in sitting 1 and get their adjustments, such as extra time or assistive tools. Sitting 2 learners are held apart and receive a different paper version, to protect assessment integrity.

Table 13. Temporary procedures, controls and resources.

Function	Temporary procedure	Approvals and controls	Records and reconciliation	Minimum staff, equipment, supplier
F1 Exam	Deliver the approved question set through the M365 assessment tool (A2). Sign-in with M365 identity, one attempt, time-limited, randomised order. Invigilate centre learners in person and own-device learners on a video call.	Head of Examinations approves the tool and the go/no-go by 09:45. Invigilator checklist; incident log for any irregularity. Privacy: access restricted to examination staff; no papers or answers on personal devices.	Responses exported to a restricted folder. After restoration, import to LMS01 and check a sample of 10% against the originals. Delete temporary copies after sign-off.	5 invigilators, 2 coordinators, 1 technician on call; 25 workstations, hotspots, power banks; alternate facility, Microsoft
F2 Fees	Record each receipt in a restricted register using the payment provider's transaction report. No wallet or ledger edits. Confirm fees by 16:00 from the provider report.	Two people per entry (record, approve). Unique provider reference prevents duplicates.	On FIN01 restoration, reconcile the register to the provider report and to FIN01 totals; investigate every difference.	2 finance officers, 1 approver; payment provider portal
F3 Admissions	Receive applications and registration changes through an online form with restricted access, open until 18:00.	Admissions Manager approves any extension; Registrar approves a deadline change. Only admissions staff see the list.	Import to REG01 after restoration; remove duplicates by applicant ID and email; confirm counts.	2 admissions staff; Microsoft
F4 Payroll	Freeze changes except urgent, dual-approved ones on a signed form. Build the payroll file from HR01 once restored; otherwise from the last approved run plus approved changes.	HR Manager and Finance Manager both approve. One bank submission only, by 12:00 Day 1, after a duplicate check.	Compare totals with the last run; log every manual change; correct any error through the bank within one cycle.	HR Manager, 1 finance approver; bank channel
F5 Records	Log requests in a controlled queue. Issue certificates only after FS01 is restored and the Registrar has verified the file.	Registrar signs urgent requests. Access follows AUTH01 groups; no records by personal email.	Match issued items to the queue; no request older than 48 hours.	2 records officers; offsite backup provider

4.4 DR runbook

The runbook follows the drill sequence. After each restore a snapshot is taken **before** acceptance, so rollback is possible. A step is finished only when its completion check passes. Restoring technology (a check on the system) is separate from resuming the business service (acceptance by the function owner).

Table 14. DR runbook.

#	Action	Prerequisite	Owner	Completion check	Contingency if the step fails
1	Activate the plan; assign technicians T1 and T2; open the decision log	Director decision	BCDRC	Log open; roles confirmed by call	Deputy decides; start without confirmation if 30 min pass
2	Build the recovery network (30 min)	None	TRL (T1)	Connectivity, DNS and firewall tested; mobile-data link tested	Use the alternate facility network; call the ISP
3	Restore AUTH01 from B03 (30 min)	Step 2	TRL (T1)	Directory starts; test user and administrator log on with the separate backup-admin identity	Use the previous snapshot; use M365 identity for exam access meanwhile

#	Action	Prerequisite	Owner	Completion check	Contingency if the step fails
4	Security and backup checks (30 min): verify checksums and chain (B01 plus incrementals), malware scan, review backup-admin logons; select the restore point	Step 3	TRL (T1)	Chain complete from B01 to the latest point; no unexplained administrator activity	Fall back to the last complete point (04:00 or B01); if activity is unexplained, stop and call the IRL
5	Restore DB01 (90 min); snapshot	Step 4	TRL (T1)	Consistency check; row counts and control totals match the last report	Restore the next-oldest point; last resort is the 17:00 Day −3 full with manual reconciliation
6	Restore FS01 (60 min) in parallel; snapshot	Steps 3 and 4	T2	Files open; permissions match AUTH01 groups	Restore the previous full backup; keep manual F5 queue
7	Restore LMS01 (45 min); snapshot	Step 5	T1	Login works; test submission saved and read back	Keep M365 assessment tool; rebuild from the application image
8	Restore REG01 (30 min) and HR01 (45 min); snapshot	Step 5	T2	Test application accepted; payroll totals match the last run	Keep the manual form and the last approved payroll file
9	Restore FIN01 (30 min); snapshot	Step 5	T1	Test receipt posts; ledger agrees with provider report to 07:00	Keep the manual register
10	Business acceptance by each function owner	Steps 6–9	FO (BCDRC coordinates)	Written sign-off against the Table 4 service level	Stay on the manual procedure; fix and retest
11	Rollback if acceptance fails or a new compromise appears	Snapshot	TRL	Service reverts to the snapshot or manual procedure; decision logged	Isolate the system and escalate to the IRL
12	Declare recovery and update stakeholders	Step 10	BCDRC	Declaration time and function status in the log and report	Delay declaration until sign-off

4.5 Communications, incident response and return to site

Table 15. Stakeholder communication matrix.

Stakeholder	Trigger	Message owner	Channel (primary / alternative)	Approver	Frequency
Learners	Activation; any change to exam or services	COM	SMS and M365 email / phone call to the exam cohort, public status page	Director	First notice 09:00; then 10:15 and every 4 h
Staff	Activation; instructions	COM	SMS and call tree / M365 Teams	BC/DR Coordinator	Every 4 h and at shift change
Director and deputy	Activation; decision needed	BCDRC	Call / SMS	–	Situation report every 2 h
Alternate facility	Activation; extension	Facilities Manager	Phone / email	Director	On activation and daily
Payment provider	Finance restoration or mismatch	Finance Manager	Provider support line / account manager	Finance Manager	On trigger
Offsite backup provider	Retrieval request; integrity failure	Technical Recovery Lead	Support line / account manager	BC/DR Coordinator	On trigger
Bank (payroll)	Payroll submission or change	HR Manager	Bank channel / phone	Director	On trigger
IR team	Possible compromise	BCDRC	Phone / secure channel	Director	Immediately
Data-protection regulator and legal adviser	Possible personal-data breach (via IRL and legal adviser)	Legal adviser	Official route / phone	Director	As advised
Media	Enquiry	COM	Holding statement	Director	Once, then on approval

Initial learner notice (141 words; required 120–180).

> Notice to learners – Monday, 09:00 WAT
>
> Flooding and a power failure have closed our main site. No one has been hurt. The site is expected to stay closed for about 72 hours.
>
> **Your examination today:** it will go ahead, using a temporary online arrangement approved by the Head of Examinations. If you need a centre workstation, we will text you by 09:30 with your sitting: 25 learners sit at 11:00 at our alternate facility, which opens at 10:00, and the others sit at 12:30. If you are using your own device, charge it, test your Northbridge Microsoft 365 sign-in now, and join at 10:45.
>
> **Support:** tell us about any accessibility or connection need at examsupport@northbridge.example or +234 000 000 0000 (fictional details).
>
> **Safety:** we will never ask for your password. Trust only official messages from us.
>
> Next update: 10:15 WAT.

When possible compromise brings in incident response. No cyberattack is confirmed, but any of these triggers an immediate call to the IRL: an unexpected backup-administrator logon, a failed integrity check, ransom or encryption indicators, unexplained changes in restored data, unusual M365 sign-ins, or provider records that disagree with restored records in a way that suggests tampering. I then pause the affected restore step, preserve logs and the backup in question, and keep business continuity running on manual procedures while the IRL leads the investigation (A11).

Safe return to the primary site. Return needs written Facilities clearance (structure, dry, electrical safety, stable power for 24 hours, network tested). I freeze changes at a planned cut-over, take a final backup at the recovery site, restore it at the primary site, and move the manual records through a controlled import with counts and control totals checked by each function owner. A day of parallel running follows. Final sign-off by the Director comes only after both Facilities clearance and business validation, and the rollback is to stay at the alternate facility.

Task 5 – Recovery schedule and exercise (15 marks)

5.1 Baseline recovery schedule (modelled)

> **MODELLED BASELINE, NOT A LIVE TEST.** The schedule below applies the supplied drill timings and assumes every backup passes its integrity check. No restoration was performed.

The schedule starts at 08:00 with two technicians (T1, T2). An action takes one technician for its whole duration, cannot be split, and starts only when its prerequisites are complete. The recovery network, AUTH01 and the security checks run in sequence on T1, because each depends on the one before. FS01 needs only AUTH01 and the checks, so T2 restores it while T1 restores DB01, which uses time that would otherwise be idle. After DB01, the four applications total 45 + 30 + 30 + 45 = 150 minutes of work, so two technicians need at least 75 minutes. I order LMS01 first, as the highest priority.

Table 16. Baseline schedule, current warm-site timings.

Action	Technician	Start	Finish	Minutes	Prerequisite
Recovery network	T1	08:00	08:30	30	None
AUTH01 restoration	T1	08:30	09:00	30	Recovery network
Security and backup checks	T1	09:00	09:30	30	AUTH01 restoration
DB01 restoration	T1	09:30	11:00	90	Security and backup checks
FS01 restoration (F5)	T2	09:30	10:30	60	AUTH01 and security checks
LMS01 restoration (F1)	T1	11:00	11:45	45	DB01
REG01 restoration (F3)	T2	11:00	11:30	30	DB01
HR01 restoration (F4)	T2	11:30	12:15	45	DB01
FIN01 restoration (F2)	T1	11:45	12:15	30	DB01

Critical path. Recovery network (30) → AUTH01 (30) → security and backup checks (30) → DB01 (90) → LMS01 (45) = 225 minutes, so **the earliest assessment service is 11:45**, 45 minutes after the examination starts. All five functions are complete by **12:15**, and no schedule can finish earlier: 11:00 plus 75 minutes. The other path (FS01) has 90 minutes of slack.

Table 17. Elapsed recovery time and comparison with objectives (modelled; elapsed minutes from 08:00).

Function	Restored	Elapsed	RTO	Margin to RTO	MTD	Restored checkpoint age	RPO	RPO result
F1 (LMS01)	11:45	225 (3 h 45)	240	15	480	1 h	1 h	Met, no margin
F2 (FIN01)	12:15	255 (4 h 15)	270	15	480	1 h	1 h	Met, no margin
F3 (REG01)	11:30	210 (3 h 30)	360	150	600	1 h	1 h	Met, no margin
F4 (HR01)	12:15	255 (4 h 15)	720	465	1,680	1 h	4 h	Met
F5 (FS01)	10:30	150 (2 h 30)	1,440	1,290	2,880	9 h	4 h	Not met: 5 h over

All RTOs and MTDs are met in the baseline, though F1 and F2 have only 15 minutes of margin. F5 misses its RPO by 5 hours because FS01 is restored to 23:00 (Task 2). **Workarounds are needed even though restoration succeeds:** the 11:00 examination (LMS01 arrives at 11:45); F2 receipts between 07:00 and the restoration time; F3 applications entered after 07:00; and F5 changes since 23:00. The 16:00, 18:00 and Day 1 deadlines themselves are met.

5.2 Tabletop exercise (simulation)

SIMULATION. I ran this 45-minute tabletop alone on paper, playing each role in turn: BC/DR Coordinator, Director, Technical Recovery Lead, Head of Examinations, Finance Manager, Facilities, IRL and Communications Lead. Nothing was restored and no real person acted. The results are my reasoned simulation and are **kept separate from the baseline schedule**.

Objectives: test activation, the examination workaround under capacity change, backup validation with handover to incident response, and finance reconciliation. Minutes map to the scenario times shown (a time jump at minute 40 reaches the restored state). Success criteria: a logged decision within 5 minutes of each inject; no restore from an unvalidated backup; an owner and revised action for every inject.

Table 18. Stress schedule (separate from the baseline) after the minute-30 inject.

Action	Tech.	Start	Finish	Note
Network, AUTH01, security checks	T1	08:00	09:30	Failure found at 09:20
Re-validate chain; choose 04:00 differential point	T1	09:30	10:00	Adds 30 min
Capture backup and logon logs with the IRL	T2	09:30	10:00	Evidence for IR
DB01 restoration to 04:00	T1	10:00	11:30	90 min
FS01 restoration	T2	10:00	11:00	After checks
LMS01	T1	11:30	12:15	F1: 255 min, RTO breached by 15 min
REG01	T2	11:30	12:00	F3
HR01	T2	12:00	12:45	F4
FIN01	T1	12:15	12:45	F2: 285 min, RTO breached by 15 min

Table 19. Simulated decision log (every inject).

Inject (minute, scenario time)	Decision and revised action	Owner	Rationale	Effect on service or recovery
10 – 08:30. Closure extends from 72 hours to five days	Extend the alternate facility booking to 120 hours; set a technician rota (12 hours on, 12 off) with vendor support for relief; re-plan teaching on M365; move the return decision to Day 5	Director (BCDRC prepares)	Two technicians cannot sustain 120 hours of continuous work; the warm site must outlast the outage	Baseline schedule unchanged; adds a sustainment workload and cost; F4 and F5 deadlines unaffected

Inject (minute, scenario time)	Decision and revised action	Owner	Rationale	Effect on service or recovery
20 – 09:00. Only 15 working workstations	Three sittings: 15 centre learners and the 80 own-device learners at 11:00, 15 at 12:30, 10 at 14:00; accessibility needs first; three paper versions; ask the facility to repair within the day	Head of Examinations (FO)	Preserves fairness and integrity; keeps the exam on Day 0; avoids the NGN 120,000 trigger if A2 holds	Last sitting ends 15:00 (A1); more invigilators; learner notice updated at 09:30
30 – 09:20. Incremental fails its check; unexpected backup-admin logon	Stop the DB01 start; call the IRL; re-validate the chain and restore to the 04:00 differential; capture logs; use the Task 5 stress schedule	BC/DR Coordinator (IRL accountable for the investigation)	A chain with a bad file or tampering could damage restored data; the logon may be a compromise	LMS01 12:15 and FIN01 12:45: both 15 min over RTO; data loss 4 h, 3 h over RPO; exam unaffected through the workaround
40 – 13:00. Provider shows successful payments missing from restored finance records	Freeze receipts and wallet credits; reconcile every provider payment after 04:00 to FIN01 with dual approval; post missing items by unique reference; check whether any payment before 04:00 is missing	Finance Manager	The restore point is 04:00, so a gap after it is expected; one before it would suggest tampering	F2 acceptance delayed to 15:00 (reconciliation); no double charges; if earlier payments are missing, the IRL is called again

5.3 Corrective actions

The simulation shows five weaknesses: shared administrator identity, an unproven exam workaround, a DB01 chain that depends on every incremental, a critical path with no slack, and weak finance reconciliation. Actions are dated assuming the exercise is run on 23 October 2026 (A9).

Table 20. Corrective actions.

#	Action	Priority	Owner	Target date	Measurable completion evidence
1	Separate backup and production admin identities, add MFA, and document provider access independence	High	Technical Recovery Lead	6 Nov 2026	Access review shows no shared identity; delete test against the offsite copy fails and is logged
2	Pre-stage and rehearse the M365 assessment workaround with the question set	High	Head of Examinations	13 Nov 2026	Mock exam with 105 users completed; 100% of responses exported; sign-off recorded
3	Implement the new DB01 schedule (04:00 and 12:00 differentials) with a nightly chain check	High	Technical Recovery Lead	20 Nov 2026	Seven consecutive nights of verified chains; one timed restore report
4	Shorten the critical path: pre-stage the network and AUTH01 images; time a drill to LMS01 ≤ 180 min	Medium	BC/DR Coordinator	22 Dec 2026	Timed drill log shows LMS01 within 180 min
5	Write and test a payment-provider reconciliation procedure and a daily settlement export	Medium	Finance Manager	20 Nov 2026	Dry run reconciles 100% of test payments within 2 hours
6	Obtain a signed alternate facility addendum guaranteeing 25 working workstations and a 5-day option	Medium	BC/DR Coordinator	27 Nov 2026	Signed addendum; on-site test shows 25 working machines

Task 6 – Governance and performance measures (10 marks)

6.1 Ownership, testing, maintenance and framework mapping

The BC/DR Coordinator owns and maintains the plan, the Director approves it, and function owners own their minimum-service procedures. The Registrar, the Technical Recovery Lead and the IRL review it. Testing is risk-based: backup restores most often because they are the weakest evidence today, and a full exercise before the flood season.

Table 21. Governance arrangements and testing calendar.

Element	Arrangement
Ownership and approval	Owner: BC/DR Coordinator. Approver: Director. Reviewers: Registrar, Technical Recovery Lead, IRL, function owners. Risk acceptance as in Table 10.
Testing calendar	Monthly: sample restore of DB01 and FS01, and a call-tree test. Quarterly: full DB01 and application restore and a tabletop with a rotating scenario. Six-monthly: alternate facility test (25 workstations). Yearly: full failover exercise in March, before the heavy rains.
Review triggers	After any real incident or exercise; a change in systems, suppliers, the site or the alternate facility; a failed test; an audit finding; a published flood warning; or a change to RTO, RPO or legal retention.

Element	Arrangement
Training	Induction for plan roles within 30 days; yearly role training; invigilator and coordinator briefing before each exercise; two technicians cross-trained with a vendor contact.
Version control and distribution	Version number, change log, approval signature and effective date on every copy; superseded copies archived. Controlled distribution list; printed copies at the alternate facility and with the Director; offline copies on the phones of plan roles.

The mapping below is an **educational alignment**, with no claim of conformity or certification. NIST SP 800-34 Rev. 1 sets out seven planning steps (NIST, 2010), and I map them to my outputs; ISO 22301:2019 clauses 4 to 10 are the themes I use for the management-system view (ISO, 2019; ICDFA, n.d., slide 9).

Table 22. Mapping to NIST SP 800-34 planning steps and ISO 22301 themes.

Framework element	Output in this report
NIST step 1: contingency planning policy	Scope, activation thresholds and authority (4.1); ownership (Table 21)
NIST step 2: business impact analysis	Task 1 BIA and impact calculations; Task 2 objectives
NIST step 3: identify preventive controls	Backup design, separate identities, immutable copy (Task 2)
NIST step 4: create contingency strategies	Package comparison and recommendation (Task 3)
NIST step 5: develop the contingency plan	Plan, RACI, temporary procedures and runbook (Task 4)
NIST step 6: testing, training and exercises	Schedule, tabletop and corrective actions (Task 5); calendar (Table 21)
NIST step 7: plan maintenance	Review triggers, version control (Table 21)
ISO 22301 clause 4 Context	Flood risk and interested parties (6.1 below); scope
ISO 22301 clause 5 Leadership	Director approval, roles and RACI (Table 12)
ISO 22301 clause 6 Planning	Objectives (Table 4); corrective actions (Table 20)
ISO 22301 clause 7 Support	Resources, training, communication (Tables 14 and 21)
ISO 22301 clause 8 Operation	BIA, strategies, plans and exercising (Tasks 1–5)
ISO 22301 clauses 9 and 10 Evaluation and improvement	KPIs (Table 23); corrective actions (Table 20)

Flood and climate consideration. ISO 22301:2019 Amendment 1:2024 requires the organisation to determine whether climate change is a relevant issue and notes that interested parties may have climate-related requirements (ISO, 2024; PECB, n.d.). For a flood-prone site (A8), I treat flooding as a recurring risk, not a one-off: a published flood warning triggers pre-emptive activation; the alternate facility and the offsite backup region must be outside the same flood plain; and the yearly exercise is scheduled before the rains.

6.2 KPIs and maturity

Table 23. Key performance indicators.

KPI	Formula or method	Data source	Target	Frequency	Owner
Recovery performance	Functions restored within RTO ÷ functions tested × 100; also elapsed ÷ RTO per function	Exercise and drill log	100% within RTO; elapsed ≤ 80% of RTO by month 12	Quarterly	Technical Recovery Lead
RPO attainment	Tests where restored checkpoint age ≤ RPO ÷ tests × 100	Restore test reports	100%	Quarterly	Technical Recovery Lead
Restoration-test coverage	In-scope systems (7) with a successful full restore in the last 90 days ÷ 7 × 100	Test register	100% within 6 months	Monthly	BC/DR Coordinator
Corrective-action closure	Actions closed by target date ÷ actions due × 100	Action tracker	≥ 90%; none overdue by more than 30 days	Monthly	BC/DR Coordinator
Training	Plan roles trained and exercised in the last 12 months ÷ plan roles × 100	Training record	100%	Quarterly	Registrar
Backup verification	Backups verified ÷ backups scheduled × 100	Backup logs	≥ 99%	Weekly	Technical Recovery Lead

Maturity. I use the five-level maturity model on slide 13 (1 initial, 2 developing, 3 defined, 4 managed, 5 optimised) (ICDFA, n.d., slide 13). I assess Northbridge at **level 2, developing**: roles, objectives and a drill exist, but restores are

untested in 90 days, the administrator identity is shared, the exam workaround is unproven and RTOs are not validated at the alternate site. A realistic 12-month target is **level 3, defined**: restore tests for all seven systems, separate identities, a rehearsed workaround, actions closed on time and quarterly KPI reporting. Level 4 needs a year of measured results.

Authorship and AI declaration

Authorship. This is my own individual submission. All scenario data are fictional. I am ready to explain any decision in this report if the instructor asks.

AI assistance. I used AI (Claude, by Anthropic) to help me understand each task, to structure the assignment and for guidance. I reviewed, edited and take responsibility for the analysis and decisions. The simulated results are fictional.

References

Style: author–date (Harvard-style), used consistently. The last column links each source to the decisions it supports.

Source	Supports (decisions in this report)
ICDFA (n.d.) Business Continuity and Disaster Recovery. GRC205 Week 21 teaching deck, slides 1–15. International Cybersecurity and Digital Forensics Academy.	BIA, RTO, RPO, MTD and strategy concepts (Tasks 1–3); maturity model, slide 13 (Task 6); ISO 22301 themes, slide 9 (Task 6).
ICDFA (n.d.) Incident Response Planning. GRC105 (course code GRC205), Week 20 teaching deck, slides 1–18, especially slide 18 (Week 21 preview: BIA, RTO, RPO, continuity and IR integration).	Link between continuity and incident response (Task 4).
NIST (2010) SP 800-34 Revision 1: Contingency Planning Guide for Federal Information Systems. National Institute of Standards and Technology, May 2010. Available at: https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-34r1.pdf	BIA approach and the seven planning steps used in the mapping (Tasks 1, 4, 6).
ISO (2019) ISO 22301:2019 Security and resilience – Business continuity management systems – Requirements. International Organization for Standardization. (Overview and preview used; the full standard was not purchased.)	Clauses 4–10 themes in the mapping (Task 6).
ISO (2024) ISO 22301:2019/Amd 1:2024 Climate action changes. International Organization for Standardization, February 2024. Available at: https://www.iso.org/standard/88412.html	Flood and climate consideration (Task 6).
PECB (n.d.) ISO introduces climate action amendments to management systems standards. Available at: https://europe.pecb.com/article/iso-introduces-climate-action-amendments-to-management-systems-standards (secondary summary of the amendment wording).	Wording of the clause 4.1 and 4.2 changes (Task 6). Confirm against the amendment text.
CISA, MS-ISAC, NSA and FBI (2023) #StopRansomware Guide. Cybersecurity and Infrastructure Security Agency, September 2023. Available at: https://www.cisa.gov/stopransomware/ransomware-guide	Offline or immutable backups, scanning backups before restoration, and testing (Task 2).
Umejesi, F. (2026) Incident Response Plan for Northbridge Skills Institute. GRC205 Week 20 assignment (own earlier work).	IR role and escalation used in Task 4 (assumption A11).